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Phenology of the Odonata assemblage in a Mediterranean stream in the north-eastern Iberian Peninsula

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Abstract. We investigated the Odonata species composition of larvae, final instar (F-0) exuviae, and adults occurring in a sequence of two different mesohabitats – pool and riffle – in a short stretch of a Mediterranean stream running down the Catalan Pre-Coastal Range (north-eastern Iberian Peninsula). We sampled them during the period 2017–2018 and recorded adults of 26 taxa and confirmed by the presence of final instar exuviae that at least 21 were breeding successfully in that environment. We describe, for the first time for this region and habitat type, larval growth patterns, timings of adult emergence, and flight periods of the most abundant species and characterize the life cycles of some of them. These life cycles can be classified into three types: i) univoltine with overwintering in the larval stage and emergence taking place in spring or summer of the following year (*Coenagrion mercuriale*, *C. puella*, *Ceriagrion tenellum*, and *Pyrrosoma nymphula*); ii) predominantly univoltine of the previous type with a small fraction of the population being semivoltine (*Anax imperator* and *Orthetrum coerulescens*, tentative conclusion); and iii) univoltine with overwintering in the egg stage, a short larval growth period in the following spring, emergence in early summer, a long maturation period, and reproduction postponed until late summer or early autumn (*Chalcolestes viridis* and *Sympetrum striolatum*). Our results may contribute to fill in gaps in the knowledge of voltinism and phenology of the life cycles of Odonata in correlation with latitude, geographical area, and habitat type.

Further key words. Dragonfly, damselfly, dragonfly community, larval growth, timings of adult emergence, flight periods, life cycle, voltinism

Introduction

Streams and rivers are dynamic ecosystems with variable spatial and temporal conditions, and this variability determines the biocenosis they can support (ALLAN & CASTILLO 2007). Notable cases are the Mediterranean streams in which abiotic factors – mainly the seasonal events of flooding

and drying over an annual cycle and its impact on the physical, chemical, and biological characteristics of the water and on the riparian area in general – set a limit on the diversity of organisms they can harbour and condition their life cycles (WILLIAMS 1996, 2006).

Aquatic invertebrates' life cycles, and in particular larval development and growth rates, are mainly influenced by temperature and photoperiod (WAGNER 1990; CORKUM & HANES 1991). This results, in general, in a decrease of voltinism with latitude and in tropical-centred taxa having more generations per year than those centred in temperate latitudes as shown by CORBET *et al.* (2006) in a review on Odonata. However, growth rates have also been reported to be conditioned by hydroperiod, being relatively higher in temporary water species (JOHANSSON & SUHLING 2004) in order to complete their development before the habitats dry up.

The western Mediterranean basin is also subjected to intense anthropic pressure – water extraction, flow regulation, increased salinity, and pollution – causing the regression or degradation of many riverine habitats (GASITH & RESH 1999; HAFIANE *et al.* 2016; KORBAA *et al.* 2018). In addition, there are environmental and climatic factors, such as fires and global warming, that have negative consequences on their conservation as well (RISERVATO *et al.* 2009). Climate change scenarios for the Mediterranean region predict that warming will be greater than the global average. Precipitation will decrease and temperature will increase producing more frequent and intense droughts. Species will face a trade-off between adaptation to changing conditions and migration to new habitats (HICKLING *et al.* 2005; SHAH *et al.* 2012; BONADA & RESH 2013). Even though some of the life history traits of the organisms inhabiting Mediterranean streams and rivers might allow them to adapt to global climate change and the associated increase in seasonality to a certain degree (FILIPE *et al.* 2013), research on dragonflies in different biogeographical regions indicates that they are sensitive to these alterations and that monitoring their communities may help provide early warning and detailed evidence of changes in the weather patterns (HASSALL & THOMPSON 2008; OTT 2010; MCCAULEY *et al.* 2018).

However, little is known about the impacts of global climate change on certain aspects of odonate biology, such as life cycle phenology and voltinism. According to some authors (*e.g.*, OERTLI 2008), temperature rise pro-

jected by climate models will likely induce adjustments in their life cycle patterns, *e.g.*, increase in larval growth rates, shortening of larval life, advancement of the onset of emergence, and lengthening of the flight period (see HASSALL *et al.* 2007). It would be reasonable, then, to expect a general increase in the voltinism of the Odonata assemblages of the Mediterranean habitats in the coming decades (JOHANSSON & SUHLING 2004; BRAUNE *et al.* 2008). In this context, information on life cycles in these dragonfly communities is needed. The present work is intended as a contribution to the study of the phenology and voltinism of the odonates of Mediterranean-climate streams in the northeast of the Iberian Peninsula, a poorly investigated area of the western Mediterranean basin. In line with these goals, we sampled lentic and lotic mesohabitats and examined species composition and patterns of larval growth, emergence timings, and flight periods of the most representative taxa.

Material and methods

Study area

The study was performed in the Sant Llorenç del Munt and Serra de l'Obac massif (north-eastern Iberian Peninsula, Lat. 41°N, Long. 2°E; Fig. 1), which belongs to the Catalan Pre-Coastal Range. The climate is typically Mediterranean with irregular rainfall (annual avg. between 500 and 800 mm, increasing with elevation) mostly concentrated in spring and autumn, while winters and summers are usually dry and very dry, respectively. Yearly average temperature is 15°C in low lying areas but can be much lower up in the mountains (DIPUTACIÓ BARCELONA 2019). Karst conditions with highly permeable substrates rich in carbonated conglomerates are common in the massif and, therefore, infiltration and subterranean drainage limit surface water flow, which may decrease shortly after the rain ceases. There are, however, several more or less permanent streams located at low altitudes, mostly derived from springs draining the aquifer (BONADA *et al.* 2007).

Study site

The studied watercourse, Sanana stream, belongs to the Llobregat river catchment and drains a small area of approximately 15 km². We have not found published information on its annual average discharge but, based

on data from similar watercourses in the vicinity, it may probably be up to 3 km³ (KAZNER et al. 2012). On the whole, the stream can be considered intermittent with pools in the no-flow period as defined in the MIRAGE project (PRAT et al. 2014). Notwithstanding this, there are some stretches, like the sampled one, which retain surface water flow even in the driest years, sometimes reduced to no more than a trickle (pers. obs. during the period 2008 to 2018). This 370 m long stretch, located in the vicinity of Can Còdol (UTM 31N/ETRS89 409250 4608389), 1.5 km from the village of Rellinars, and 30 km from the city of Barcelona, is at an elevation of approximately 310 m a.s.l. and has an average slope of 6.8 %. At this location, the stream bed was enclosed between relatively steep slopes, which reduced total insolation, especially during the coldest months of the year. The meteorological database of the nearby Rellinars weather station indicates that, at the time the field work was done (2017–2018), the average yearly air temperature was 14°C (monthly avg.: min. 4.1/4.5°C [i-2017/ii-2018]; max. 23.5/23.4°C [viii-2017/vii–viii-2018]) and the mean annual precipitation was 485.5 mm in 2017 and 952.7 in 2018 with events of 74.9 mm/day in March 2017 and 69 in October 2018. The wettest months were March 2017 (126 mm) and October 2018 (178.3 mm) and the driest were December 2017 and 2018 with 7.5 and 5.1 mm, respectively. Vegetation adjacent to the watercourse was dominated by evergreen oaks (*Quercus ilex* L.), Aleppo pines (*Pinus*

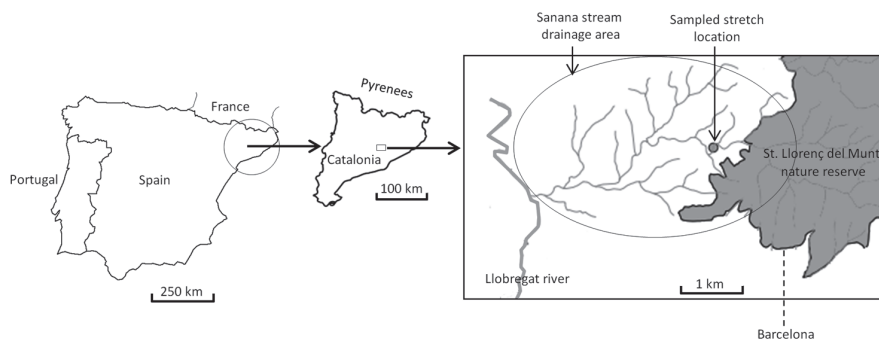


Figure 1. Location of the Sanana stream and the sampled stretch near Can Còdol in the Llobregat river catchment, in the north-eastern Iberian Peninsula. Maps obtained from the Catalan Water Agency (Agència Catalana de l'Aigua; <http://aca.gencat.cat/>) and the Barcelona Provincial Council (DIPUTACIÓ BARCELONA 2019).

halepensis Miller), and sclerophyllous shrubs except on rock outcrops, and riparian vegetation was scarce for lack of enough soil to hold it. Helophytes (*Phragmites australis* (Cav.), *Typha latifolia* L., *Carex* L., *Equisetum* L.) and hydrophytes (*Chara* L., filamentous algae) were abundant both in the riffles and pools and on the banks but varied from year to year due to floods that washed them away to varying extents.

Data collection and analysis

During 35 visits from mid-April to mid-November 2017, we searched the stream stretch weekly and collected last instar exuviae (F-0) along the margins of a number of consecutive sections ranging from 17 to 63 m in length and constituting a riffle-pool sequence.

On each visit, there was no fixed time for collection. We proceeded until no more exuviae could be found, spending an average of 120 min. We preserved the specimens in plastic containers and counted and identified them, if possible, to species and sex with a stereomicroscope. The identification keys used were those of JÖDICKE (1994), HEIDEMANN & SEIDENBUSCH (2002), and BROCHARD et al. (2012). We characterized the pattern of emergence using Westermann's criteria (WESTERMANN 2002), the *peak of emergence* being the time interval between the first and third quartiles of all the exuviae found and the *main period of emergence* that between the 10th and 90th percentiles.

On the same visit days, we searched each section for adult dragonflies of both sexes for a 10-minute period. Observation always took place between 11:00 and 18:00 h CEST (UTC +2; in November CET, UTC +1) under warm, sunny, and windless conditions with air temperatures $\geq 17^{\circ}\text{C}$ and at least 50% sunshine (BROOKS 1993). At the riffles, we walked along the banks identifying and recording all adult individuals seen within a fixed distance of 2.5 m on either side of the channel. At the pools, we walked the perimeter and recorded all individuals seen over the water and around the margin to a distance of at least 5 m including the surrounding herbaceous and shrubby growth. The flight period of each species was established considering the first and last observations for the combined sections.

We sampled larval odonates systematically on a monthly basis in two representative sections – one pool and one riffle (Fig. 2) – from September

2017 to September 2018. Sampling with a hand-net (30×25 cm, 1 mm meshed) consisted of five 1.5 m sweeps at the pool and two (upstream) at the riffle in various microhabitats taking into account the different types of concealment among the species (weed-dwelling, burrowing...). This method provides a good picture of the invertebrate fauna in this kind of watercourse, a substantial part of which is choked with vegetation (CASTELLA 1987). The contents of the net were sorted in the field until all larvae were withdrawn. We identified the collected larvae to species – except for the earliest instars – and measured them for head width, including the eyes, to an accuracy of 0.05 mm under a stereoscopic microscope using a calibrated



Figure 2. Mesohabitat types – riffle (left) and pool (right) – within the selected Sanana stream stretch near Can Còdol in the Llobregat river catchment, in the north-eastern Iberian Peninsula. The sections shown here were the sampling sites for larvae. The pool is located immediately downstream of the riffle. The riffle section is 30 m long and, during average flow conditions, has a braided channel of moderate depth (a few centimetres) with the smaller channels being 0.8 m wide in average. At the time of our study, 90 % of its surface was choked with helophytes (mostly reeds, sedges, and rushes). The pool, with a surface area of approximately 130 m², had more than 60 % of its surface covered in a similar way with submerged and emergent macrophytes (mainly stoneworts and reeds).

eye-piece graticule. The keys used were CARCHINI (1983), CONESA (1985), JÖDICKE (1994), HEIDEMANN & SEIDENBUSCH (2002), CHAM (2007, 2009), and BROCHARD et al. (2012). To study larval growth patterns and development time, we grouped head-width data in classes of 0.1 mm except for *A. imperator* (0.2 mm) (see LAMELAS-LÓPEZ et al. 2017).

On all visits we measured the water temperature at each section (Fig. 3) between 10:30 and 12:30 h. On some dates we also measured water physicochemical variables, such as conductivity, dissolved oxygen, and pH at the two sections indicated in Figure 2, especially at the times of maximum and minimum water flow, using a YSI Professional Plus multimeter (Table 1).

Results

Water temperature and dissolved oxygen measurements yielded very different values at the times of minimum (26-xi-2017) and maximum (21-iv-2018) stream flow, while conductivity and pH remained relatively constant (Table 1). The water pH was slightly basic in accordance with the geological

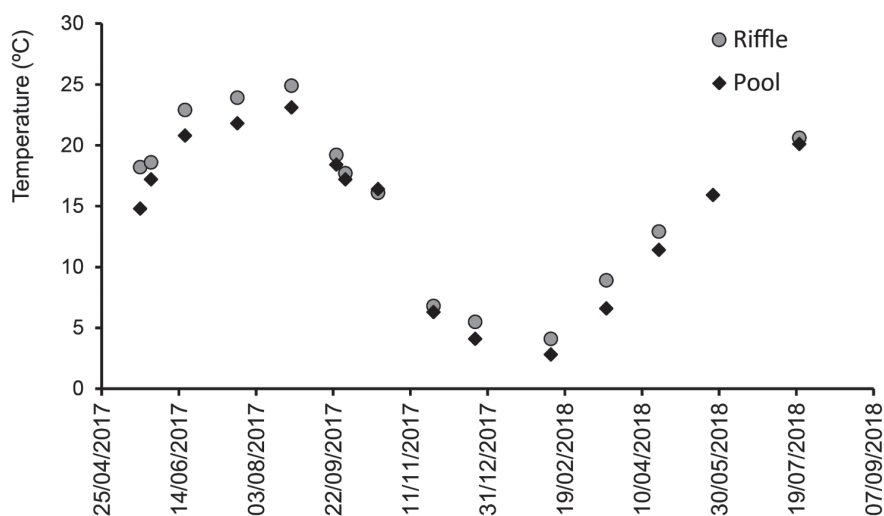


Figure 3. Water temperature measurements taken between 10:30 and 12:30 h (CEST) in the Sanana stream near Can Còdol in the Llobregat river catchment, in the north-eastern Iberian Peninsula, in the mesohabitat sections sampled for larvae (see Fig. 2) during the study period.

Table 1. Results of the water physical parameters analyses of the Sanana stream near Can Còdol in the Llobregat river catchment, in the north-eastern Iberian Peninsula, at the times of minimum (26-xi-2017) and maximum (21-iv-2018) stream flow at the riffle and pool shown in Figure 2. Measurements taken between 10:30 and 12:30 h CEST (in November CET).

Meso-habitat	Date	Time (CEST)	T _{water} [°C]	P [mm Hg]	Oxygen [%]	Oxygen [mg L ⁻¹]	Conduct. [μS cm ⁻¹]	pH
Pool	26-xi-2017	10:30	6.3	744.6	35.0	4.3	648.8	8.30
	21-iv-2018	11:50	11.9	739.8	93.3	10.1	658.6	8.49
Riffle	26-xi-2017	12:05	6.8	744.0	38.3	4.7	649.1	8.35
	21-iv-2018	12:20	12.9	739.2	92.9	9.8	659.0	8.56

substrate. In the riffle and pool shown in Figure 2, the lowest recorded water temperatures were, respectively, 4.1 and 2.8°C (ii-2018) and the highest 27.4 and 26.9°C (viii-2017) (Fig. 3).

In total, we collected and examined 975 larvae and 887 F-0 exuviae and identified 1838 adults. We recorded adults of 26 species (16 Anisoptera and 10 Zygoptera), 21 of which proved to be breeding successfully in the sampled stretch (found as F-0 exuviae), and 15 were found as larvae (Table 2). Per mesohabitat type, the number of species found as larvae or F-0 exuviae was higher in the pool sections (21 spp.) where the most abundant, ranked by total number of larvae plus F-0 exuviae collected, were *C. puella*, *C. tenellum*, *I. graellsii*, *A. imperator*, *P. nymphula*, and *C. viridis*. At the riffles (18 spp.), the most abundant were *O. coerulescens* and *C. mercuriale*. We found the larvae and F-0 exuviae of some species, like *C. mercuriale*, *O. coerulescens*, and *C. haemorrhoidalis*, almost exclusively in the riffles, while the rest were encountered primarily in the pools. However, we found most species in both mesohabitats in varying proportions (Table 2). We found neither F-0 exuviae nor larvae of four species – *S. fusca*, *A. parthenope*, *L. depressa*, and *T. kirbyi* – and therefore, we have considered them to be non-breeding or accidental. Imagines of most species congregated predominantly in the pools although those of a few – *C. haemorrhoidalis*, *C. mercuriale*, *C. boltonii*, *O. coerulescens*, and *O. brunneum* – were found mainly in the riffles.

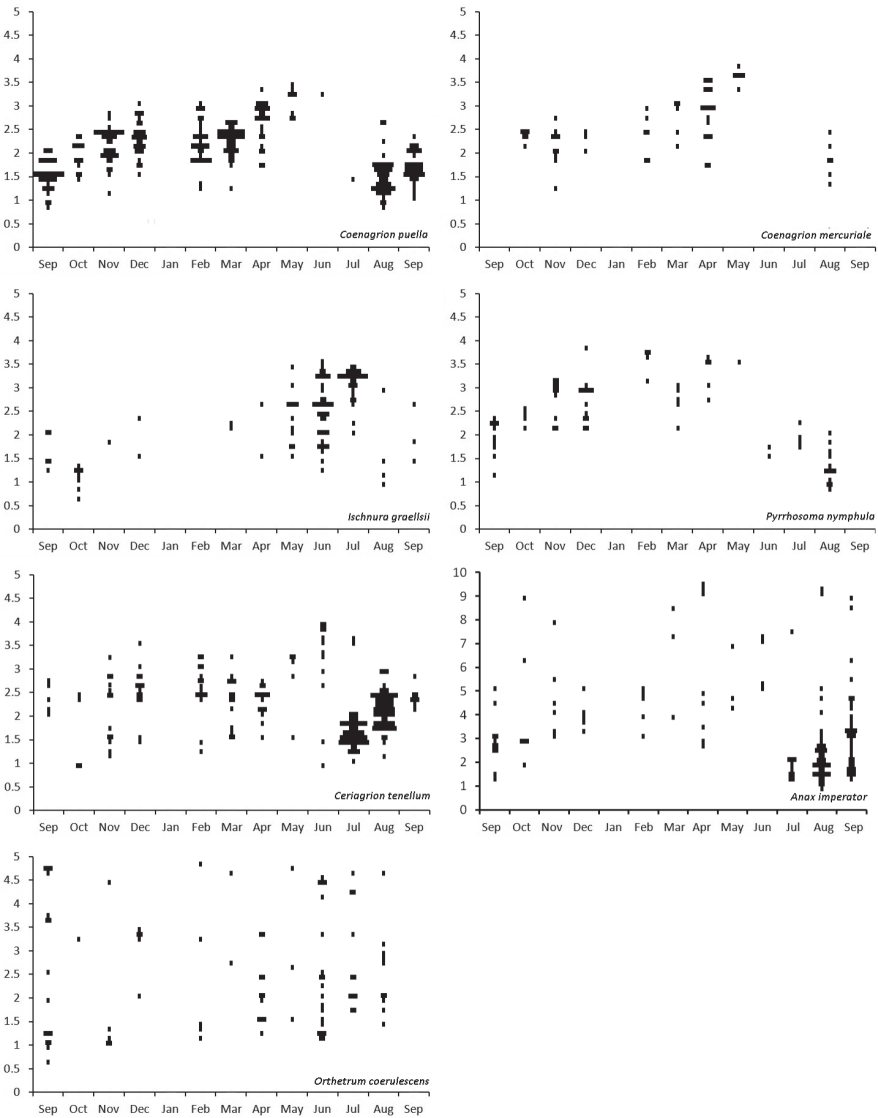


Figure 4. Head width [mm] frequency distribution obtained by monthly samplings of the Sanana stream near Can Còdol in the Llobregat river catchment, in the north-eastern Iberian Peninsula, from September 2017 to September 2018. Data were grouped in class intervals of 0.1 mm except for *A. imperator* (0.2 mm). Each small solid rectangle represents one individual. No samples were taken in January 2018.

Table 2. Species recorded at the Sanana stream near Can Còdol in the Llobregat river catchment, in the north-eastern Iberian Peninsula. Left-hand side columns: number of larvae, F-0 exuviae, and adults recorded at each mesohabitat type (R – riffle, P – pool, N – sum of all the individuals observed or collected). Middle

Taxon	Larvae			Exuviae		
	R	P	N	R	P	N
<i>Calopteryx haemorrhoidalis</i> (Vander Linden, 1825)	7	1	8	0	0	0
<i>Chalcolestes viridis</i> (Vander Linden, 1825)	0	9	9	16	74	90
<i>Sympetma fusca</i> (Vander Linden, 1820)	0	0	0	0	0	0
<i>Coenagrion caerulescens caerulescens</i> (Fonscolombe, 1838)	1	5	6	10	12	22
<i>Coenagrion mercuriale</i> (Charpentier, 1840)	59	0	59	1	1	2
<i>Coenagrion puella</i> (Linnaeus, 1758)	0	270	270	2	116	118
<i>Ceriagrion tenellum</i> (Villers, 1789)	19	196	215	1	183	184
<i>Erythromma lindenii</i> (Selys, 1840)	0	0	0	0	2	2
<i>Ischnura graellsii</i> (Rambur, 1842)	2	117	119	0	21	21
<i>Pyrrhosoma nymphula</i> (Sulzer, 1776)	16	54	70	3	88	91
<i>Platycnemis latipes</i> Rambur, 1842	0	0	0	1	0	1
<i>Gomphus pulchellus</i> Selys, 1840	2	2	4	0	46	46
<i>Onychogomphus forcipatus unguiculatus</i> (Vander Linden, 1820)	0	0	0	0	1	1
<i>Boyeria irene</i> (Fonscolombe, 1838)	0	0	0	1	2	3
<i>Aeshna cyanea</i> (Müller, 1764)	0	9	9	6	9	15
<i>Aeshna mixta</i> Latreille, 1805	0	0	0	1	4	5
<i>Anax imperator</i> Leach, 1815	1	104	105	0	121	121
<i>Anax parthenope</i> (Selys, 1839)	0	0	0	0	0	0
<i>Cordulegaster boltonii</i> (Donovan, 1807)	2	0	2	1	3	4
<i>Libellula depressa</i> Linnaeus, 1758	0	0	0	0	0	0
<i>Orthetrum coerulescens</i> (Fabricius, 1798)	82	7	89	10	12	22
<i>Orthetrum brunneum</i> (Fonscolombe, 1837)	0	0	0	4	10	14
<i>Crocothemis erythraea</i> (Brullé, 1832)	0	1	1	0	1	1
<i>Sympetrum striolatum</i> (Charpentier, 1840)	0	9	9	5	70	75
<i>Sympetrum sinaiticum</i> Dumont, 1977	0	0	0	0	49	49
<i>Trithemis kirbyi</i> (Selys, 1891)	0	0	0	0	0	0

columns: emergence patterns (main period of emergence: time interval between the 10th and 90th percentiles of all the exuviae found; peak of emergence: time between the first and third quartiles). Right-hand side column: flight periods. Larvae were sampled at the mesohabitat sections shown in Figure 2.

Adults			Emergence				Flight period
R	P	N	First exuvia	Last exuvia	Main period	Peak	
108	47	155	—	—	—	—	27-v–17-ix
38	38	76	02-vii	03-ix	02-vii–29-vii	09-vii–22-vii	24-vi–12-xi
0	10	10	—	—	—	—	—
21	41	62	—	—	—	—	27-v–29-vii
9	0	9	—	—	—	—	04-vi–9-vii
126	119	245	14-v	02-vii	14-v–04-vi	20-v–04-vi	07-v–29-vii
132	141	273	04-vi	13-viii	04-vi–2-vii	11-vi–02-vii	27-v–30-ix
2	27	29	—	—	—	—	11-vi–22-vii
56	99	155	14-v	13-viii	20-v–05-viii	18-vi–22-vii	14-v–10-ix
53	87	140	15-iv	04-vi	15-iv–20-v	23-iv–07-v	15-iv–09-vii
28	33	61	—	—	—	—	04-vi–13-viii
0	5	5	20-v	11-vi	20-v–04-vi	20-v–27-v	—
2	0	2	—	—	—	—	—
0	3	3	—	—	—	—	—
11	25	36	18-vi	10-ix	—	—	15-vii–12-xi
2	10	12	—	—	—	—	10-ix–29-x
31	60	91	20-v	20-viii	27-v–22-vii	04-vi–02-vii	07-v–26-viii
1	0	1	—	—	—	—	—
3	1	4	—	—	—	—	—
2	0	2	—	—	—	—	—
174	62	236	04-vi	10-ix	04-vi–29-vii	24-vi–22-vii	11-vi–17-ix
16	4	20	18-vi	22-vii	—	—	11-vi–26-viii
6	16	22	—	—	—	—	11-vi–26-viii
54	92	146	02-vii	03-ix	09-vii–05-viii	15-vii–29-vii	11-vi–12-xi
4	38	42	22-vii	03-ix	22-vii–05-viii	29-vii–5-viii	29-vii–29-x
0	1	1	—	—	—	—	—

Combining the results of our field research on duration of larval life, timings of emergence and flight periods (see Figs 4–6), we have been able to detail the life cycle phenology of the following eight species: *C. viridis*, *C. mercuriale*, *C. puella*, *C. tenellum*, *P. nymphula*, *A. imperator*, *O. coerulescens*, and *S. striolatum* (Fig. 7). Regarding *C. viridis* and *S. striolatum*, we did not detect any larvae belonging to these species during the samplings performed in late summer, autumn, and winter, and we concluded that all the individuals were in the egg stage. Larvae began to appear in our samples in spring and continued to do so until emergence took place in early and mid-summer. From each sample, we examined only a few individuals in order to track their development but we did not obtain abundance data.

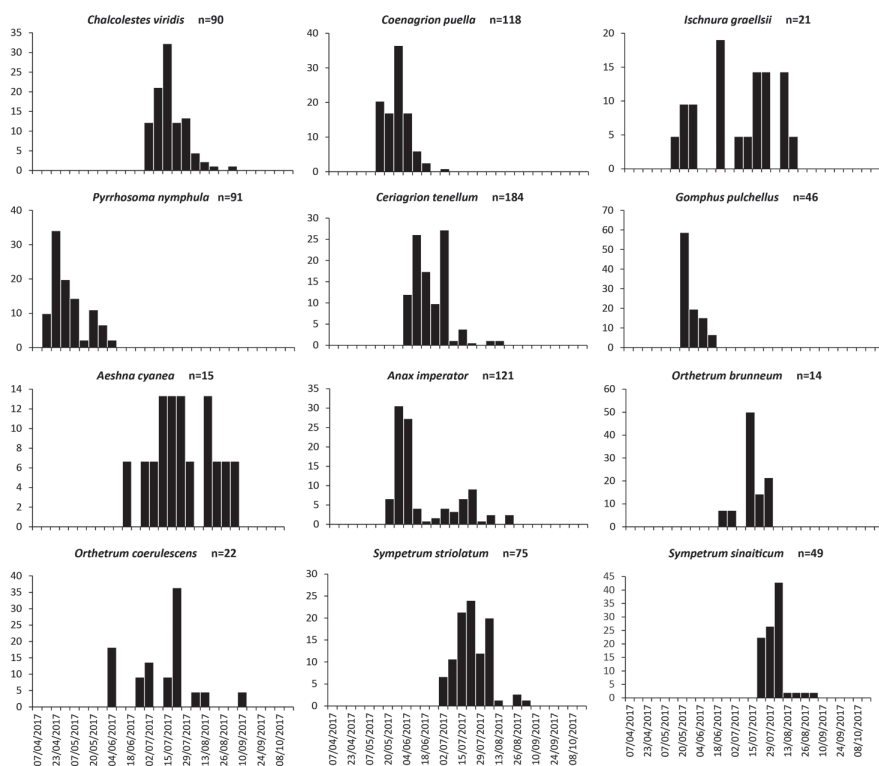


Figure 5. Percentage frequency distributions of F-0 exuviae during the emergence period based on weekly samplings of the Sanana stream near Can Còdol in the Llobregat river catchment, in the north-eastern Iberian Peninsula.

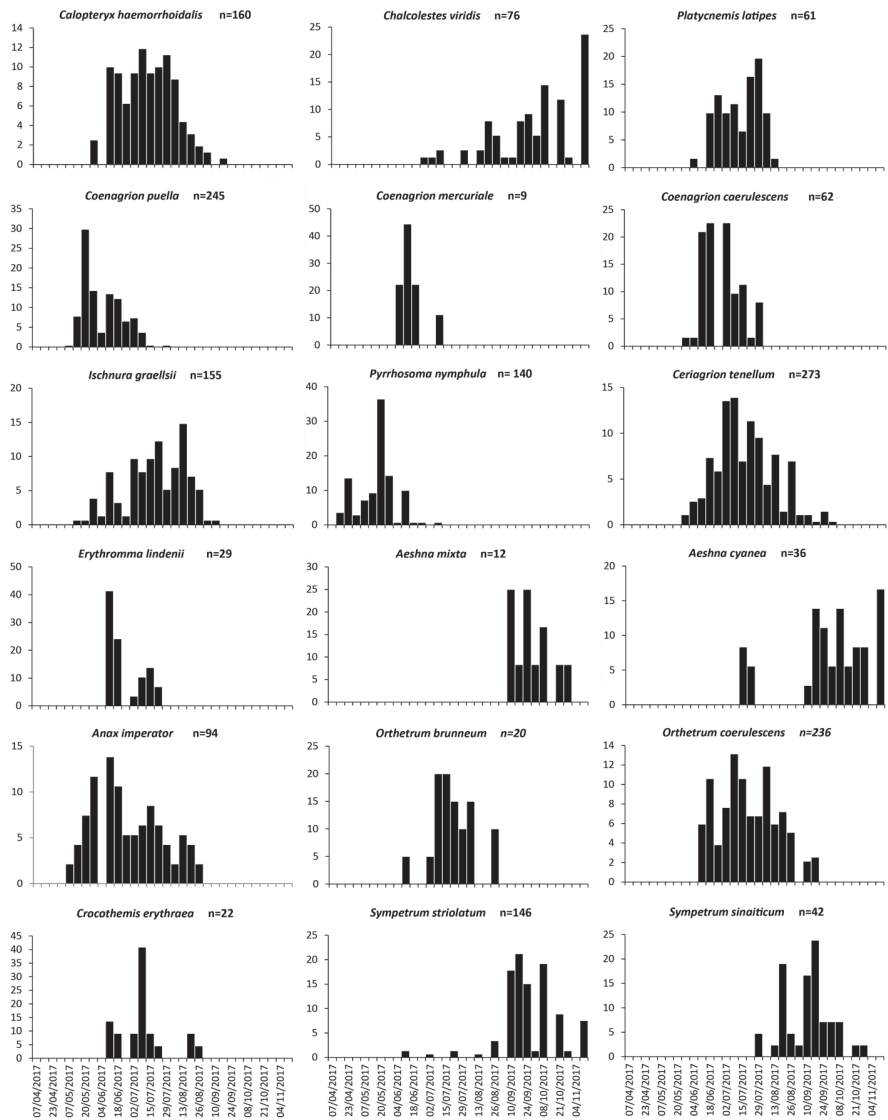


Figure 6. Percentage frequency distributions of adults during the flight season based on weekly observations at the Sanana stream near Can Còdol in the Llobregat river catchment, in the north-eastern Iberian Peninsula.

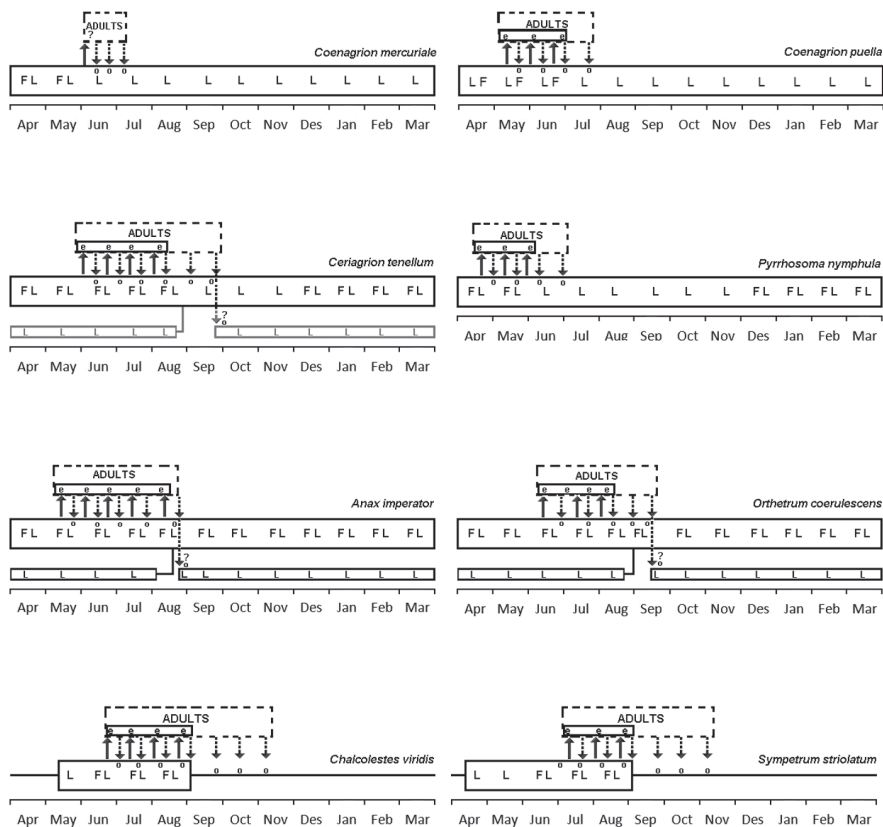


Figure 7. Life cycles of the Odonata species in the sampled mesohabitats at the Sanana stream near Can Còdol in the Llobregat river catchment, in the north-eastern Iberian Peninsula. *Coenagrion mercuriale*, *C. puella*, *Ceriagrion tenellum*, and *Pyrrhosoma nymphula*: univoltine with overwintering in the larval stage; *Anax imperator* and *Orthetrum coerulescens*: mostly univoltine of the previous type but partially semivoltine (tentative explanation), the lower thinner bars represent the life cycle phenology of the semivoltine fraction of the population (this also applies to the *C. tenellum* graph, although in this case the semivoltine portion of the population, in grey colour, would be much smaller); *Chalcolestes viridis* and *Sympetrum striolatum*: univoltine with overwintering in the egg stage. The field data used to deduce these patterns were collected in 2017 and 2018. L – larval stages except the last one; F – final larval stage; e – emergence; o – oviposition. Continuous horizontal line indicates eggs. Up-arrows indicate emergences and dashed down-arrows ovipositions. Connection line bent at a right angle from lower to middle bar indicates prolongation of larval life for an additional winter.

The life cycles of these species can be classified into three types:

- i) univoltine with slow larval growth and overwintering in the larval stage (*C. mercuriale*, *C. puella*, *C. tenellum*, and *P. nymphula*);
- ii) partially semivoltine in a predominantly univoltine population of the previous type (*A. imperator* and *O. coerulescens*). There is evidence, reflected in Figures 4–6, suggesting that these species might exhibit this kind of flexible life-history strategy although the results are difficult to interpret (see Discussion); and
- iii) univoltine with overwintering in the egg stage and fast larval growth (*C. viridis* and *S. striolatum*).

For other common species found breeding in the stream, we also obtained valuable information on the timings of one or more life history events, such as growth, emergence, and flight season (Figs 4–6).

Discussion

Although we took particular care to find the smallest larvae in each sample, we were not able to provide head width values of early larval instars in the growth graphs in Figure 4 due to difficulties involved in their capture (mesh size) and in their taxonomical identification. To be able to identify these early instars, it is usually necessary to monitor hatching and growth in the laboratory. But there, however, the environmental conditions are not easily replicable, and the results are less realistic (LAMELAS-LÓPEZ et al. 2017); but see INGRAM (1976), SCHÜTTE et al. (1997), DE BLOCK et al. (2005) and LAMBRET et al. (2017), who studied hatching in the field under natural conditions.

Notwithstanding this, we have been able to delineate the life cycle phenology and voltinism exhibited by eight of the most abundant species found breeding in the studied stream stretch (Fig. 7). As pointed out previously, we have sorted them into three groups following approximately the classification proposed by CAYROU & CÉRÉGHINO (2005) for the life-history patterns of aquatic insects:

- i) Four species have a univoltine life cycle with oviposition taking place throughout late spring and summer (April–September), slow larval growth occurring over a rather long period (10–12 months), overwintering in the larval stage, and emergence ensuing more or less synchronously in spring

or summer of the following year. This is the case for *C. mercuriale*, *C. puella*, *P. nymphula*, and *C. tenellum*. In the larval growth graphs of the first three taxa, there is a clear-cut separation between consecutive generations (Fig. 4) indicating an annual periodicity of reproduction. The emergence and flight period graphs also confirm this pattern. In the graph of *C. tenellum* the boundary between successive generations is not as well-marked, and we also find a wider range of head width sizes in the course of the year, which may be attributed to its more protracted breeding season. However, it might also be evidence of semivoltinism in a very small fraction of the population (see CORBET 1957). *Coenagrion mercuriale* has been described as semivoltine in Britain by PURSE & THOMPSON (2002) and as bivoltine in the southern end of its distribution (MAHDJOUB et al. 2015), *C. puella* has been reported to be univoltine in Austria by WARINGER & HUMPESCH (1984) and in Italy by NICOLAI & CARCHINI (1985), and *P. nymphula* as univoltine from Switzerland (ROBERT 1958) but semi- and partivoltine from Britain (BENNETT & MILL 1993).

ii) Two taxa (*A. imperator* and *O. coerulescens*) may probably have a flexible life cycle consisting of partial semivoltinism in a predominantly univoltine population of the type discussed in the previous paragraph, but the analysis of the larval growth, emergence, and flight period patterns is inconclusive. In the case of *A. imperator*, we found a large range of larval sizes throughout the year due to its protracted reproductive period. The larval growth graph (Fig. 4) suggests that a majority of individuals might have a univoltine life cycle, spending the winter as larvae and emerging during the following year. However, there is a small proportion of the *A. imperator* population that appears to be semivoltine. It most probably consists of individuals originating from eggs laid late in the season that do not have time to complete their development within the next year and spend a second winter as larvae to emerge in the following spring. The spring emergence peak, spanning from May to June (Fig. 5), would consist of a mixture of the two groups, while the second peak, occurring in summer, would include mostly the univoltine type. The possibility of bivoltinism does not seem likely if we follow this reasoning: If *A. imperator* larvae hatched from early eggs – laid at the beginning of summer – were to emerge in autumn of the same season, we would probably have detected some emergences by the end of September

or early October. This would have been in accordance with D'AGUILAR et al. (1986), who indicate a larval development time of 90 days in the Mediterranean region. However, the last emergences we noted corresponded to a few individuals on 20th August, and we recorded the last adults at the end of August. If there had been a second generation, the flight period would probably have been longer. LAMELAS-LÓPEZ et al. (2017) provide evidence of a univoltine life cycle for this species in the Azores archipelago. On the other hand, JÖDICKE (1996) suggests a bivoltine pattern in the province of Tarragona (north-eastern Iberian Peninsula) with continuous emergence in May and June, and SÁNCHEZ et al. (2009) report that in Extremadura (central Iberian Peninsula) the species can be uni- or bivoltine depending on the temperature. In the case of *O. coerulescens*, the larval growth, emergence, and flight period graphs follow a similar pattern to those of *A. imperator*, apparently indicating a basically univoltine life cycle with a small part of the population being semivoltine. According to JULLERAT (2002), there may be fast (univoltine) and slow (semivoltine) growing larvae in the populations of this species in the Swiss Jura Mountains.

NORLING (1984) proposes a model of seasonal regulation for the Odonata of temperate latitudes based on photoperiod. According to the model, larvae that reach a so-called critical size at the end of autumn will grow rapidly during the following spring and will emerge. Larvae that do not reach the critical size in late autumn enter diapause and, when they experience longer photoperiods in the following spring, start to grow again very slowly for another year and therefore postpone emergence until after a second winter diapause. They would form the semivoltine cohort. In our study, though, we could not obtain evidence of this kind of response in *A. imperator* and *O. coerulescens*. For this purpose and for describing their life cycles with certainty, we would need to collect a larger number of larvae.

iii) Two species (*C. viridis* and *S. striolatum*) have a univoltine life cycle with overwintering in the egg stage, a short larval growth period (3–4 months) during the following spring, emergence taking place in summer, a long maturation period, and postponed reproduction until late summer or early autumn. This type of life cycle is characteristic of these species in the Mediterranean region, and there are a few publications describing it. For instance,

SAMRAOUI et al. (1998) confirm a similar phenology with a protracted aestivation period for *S. striolatum* in Algeria. JÖDICKE (1996) found this long maturation and postponed flight period in populations of *C. viridis* and *S. striolatum* in Tarragona (north-eastern Iberian Peninsula) and FERRERAS-ROMERO & MÁRQUEZ-RODRÍGUEZ (2014) in the Sierra Morena Mountains (Andalusia, southern Iberian Peninsula). FERRERAS-ROMERO & GARCÍA-ROJAS (1995) and AGÜERO-PELEGRIN et al. (1999) had previously reported it for *C. viridis* from the same area. Univoltinism in these species may be conditioned by phylogenetic constraints, as for most other *Lestes* and *Sympetrum*, although there may be exceptions, such as *S. striolatum* in north-western Germany (JÖDICKE & THOMAS 1993).

Spatial and temporal variability is an essential consideration in the study of the major events in the life histories of aquatic insects, such as feeding and growth, emergence, movements, and migrations (RESH & ROSENBERG 2012). It is highly likely then, that the patterns of these events differ in space and time depending not only on water temperature and photoperiod, which are considered the two pivotal regulatory factors, but also on other meso-habitat characteristics, such as size, water physicochemistry, flow regime, macrophyte cover, and prey availability (BRAUNE et al. 2008; REMSBURG 2011). In the case of dragonflies, which are also responsive to terrestrial drivers (REMSBURG & TURNER 2009), the timing of adult emergence and the length of the reproductive period depend on air temperature (HASSALL et al. 2007). Since our sampling was limited in space and time to a stretch of the stream and to a two-year period, we were not able to study the effects of this spatio-temporal variability on the phenology of the Sanana stream Odonata assemblage. Further research in the streams of the massif is needed to answer the uncertainties and questions raised by our results.

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